Physical Cume
November 10, 2006

A particle of mass m is in a one-dimensional box of width a. At time t=0 the particle is in the state

$$\psi(x, 0) = (3\phi_2 + 4\phi_9)/\sqrt{(25)}$$

The ϕ_n functions are the orthonormal eigenstates of the Hamiltonian operator:

$$\phi_n = \sqrt{(2/a)}\, \sin(n\pi x/a)$$

Questions:

1.) Determine whether $\psi(x, 0)$ is normalized and calculate the normalization constant, if it is not.

2.) For the first ten states, n=1 to n=10, calculate the probability $P(E_n)$ that measurement of energy, E, at t=0 finds the value E_n for the system in the state $\psi(x, 0)$ given above.

3.) In an ensemble of 2500 identical one-dimensional boxes, each containing an identical particle in the same state $\psi(x, 0)$ given above, approximately how many will be found, when E is measured at t=0,

a) in the ground state; b) in E_2; c) in E_9 ?

4.) Is there a chance that, in an ensemble of 10^{17} boxes, measurement of E finds E_2 in all 10^{17} boxes?